

LOGICAL REASONING - MCQ PRACTICE PAPER

Mid-Term Examination | Total Marks: 25 | Time: 30 Minutes
Each question carries 1 mark. Choose the most appropriate answer.

NUMBERS (3 marks)

Q1. What is the sum of the first 20 odd natural numbers?

- a) 200
- b) 400
- c) 380
- d) 420

Q2. The sum of two numbers is 40. What is the maximum possible product?

- a) 396
- b) 400
- c) 380
- d) 420

Q3. How many factors does 360 have?

- a) 20
- b) 24
- c) 18
- d) 12

PERCENTAGE (3 marks)

Q4. If the price of a commodity increases by 25%, by what percent must consumption be reduced to keep expenditure the same?

- a) 25%
- b) 20%
- c) 30%
- d) 15%

Q5. A number is first increased by 20% and then decreased by 20%. The net change is:

- a) No change
- b) 4% decrease
- c) 4% increase
- d) 2% decrease

Q6. Ram's salary is 30% less than Shyam's. By how much percent is Shyam's salary more than Ram's?

- a) 30%
- b) 42.86%
- c) 40%
- d) 35%

PROFIT AND LOSS (3 marks)

Q7. A shopkeeper buys an article for Rs. 500 and marks it at Rs. 800. He gives a 10% discount. His profit percentage is:

- a) 40%
- b) 44%
- c) 48%
- d) 36%

Q8. Two articles are sold at Rs. 5000 each. One at 25% profit, other at 25% loss. The overall result is:

- a) No profit, No loss
- b) 6.25% loss
- c) 6.25% profit
- d) 5% loss

Q9. A dishonest shopkeeper sells at cost price but uses 800 gm weight instead of 1 kg. His profit is:

- a) 20%
- b) 25%
- c) 30%
- d) 15%

AVERAGE (3 marks)

Q10. The average of first 50 natural numbers is:

- a) 25
- b) 25.5
- c) 26
- d) 26.5

Q11. The average marks of 30 students was 72. One student's marks were read as 85 instead of 55. The correct average is:

- a) 71
- b) 70
- c) 73
- d) 72.5

Q12. The average weight of 16 boys is 50.25 kg and that of 8 boys is 45.15 kg. The average weight of all 24 boys is:

- a) 47.55 kg
- b) 48.55 kg
- c) 49.25 kg
- d) 46.70 kg

PERMUTATION & COMBINATION (4 marks)

Q13. In how many ways can 6 people be seated around a circular table?

- a) 720
- b) 120
- c) 60
- d) 360

Q14. How many arrangements can be made from the letters of LEADER?

- a) 720
- b) 360
- c) 180
- d) 240

Q15. From 7 men and 4 women, a committee of 5 is to be formed with at least 1 woman. Number of ways:

- a) 441
- b) 462
- c) 420
- d) 480

Q16. In how many ways can the letters of MISSISSIPPI be arranged?

- a) 34650
- b) 36540
- c) 34560

d) 35640

PROBABILITY (4 marks)

Q17. Two dice are thrown. Probability of getting a sum of 7:

- a) $1/6$
- b) $5/36$
- c) $7/36$
- d) $1/9$

Q18. A card drawn from a deck of 52 cards. Probability it is a king or a queen:

- a) $1/13$
- b) $2/13$
- c) $3/13$
- d) $4/13$

Q19. Three students' chances of solving a problem are $1/2$, $1/3$, and $1/4$. The probability the problem is solved:

- a) $1/4$
- b) $1/24$
- c) $23/24$
- d) $3/4$

Q20. A leap year has 53 Sundays. The probability is:

- a) $1/7$
- b) $2/7$
- c) $3/7$
- d) $4/7$

WORK & CHAIN RULE (2 marks)

Q21. A can do a work in 10 days, B in 15 days. Together they complete it in:

- a) 5 days
- b) 6 days
- c) 7 days
- d) 8 days

Q22. If 12 men complete a work in 20 days, 15 men will complete it in:

- a) 14 days
- b) 16 days
- c) 18 days
- d) 25 days

SERIES (2 marks)

Q23. Find the next term: 2, 6, 12, 20, 30, ?

- a) 40
- b) 42
- c) 36
- d) 48

Q24. Find the missing term: B, D, G, K, P, ?

- a) T
- b) U
- c) V
- d) W

CODING - DECODING (1 mark)

Q25. If in a certain code COMPUTER is written as DPNQVUFS, how is PRINTER written in that code?

- a) QSJOUFR
- b) QSJOUFS
- c) QSJOUES
- d) RSJOUFR

ANSWER KEY WITH EXPLANATIONS

Q.No	Answer	Q.No	Answer
Q1	b) 400	Q2	b) 400
Q3	b) 24	Q4	b) 20%
Q5	b) 4% decrease	Q6	b) 42.86%
Q7	b) 44%	Q8	b) 6.25% loss
Q9	b) 25%	Q10	b) 25.5
Q11	a) 71	Q12	b) 48.55 kg
Q13	b) 120	Q14	b) 360
Q15	a) 441	Q16	a) 34650
Q17	a) 1/6	Q18	b) 2/13
Q19	d) 3/4	Q20	b) 2/7
Q21	b) 6 days	Q22	b) 16 days
Q23	b) 42	Q24	c) V
Q25	b) QJSJOUFS		

Detailed Explanations:

Q1. Sum of first N odd natural numbers = $N^2 = 20^2 = 400$

Q2. Maximum product when numbers are equal = $(40/2)^2 = 20^2 = 400$

Q3. $360 = 2^3 \times 3^2 \times 5^1$. Number of factors = $(3+1)(2+1)(1+1) = 24$

Q4. Reduction = $100 \times R/(100+R) = 100 \times 25/125 = 20\%$

Q5. Net change for same % increase then decrease = $-R^2/100 = -400/100 = -4\%$

Q6. $100 \times R/(100-R) = 100 \times 30/70 = 42.86\%$

Q7. SP = $800 \times 90/100 = \text{Rs. } 720$. Profit = $720 - 500 = 220$. Profit% = $(220/500) \times 100 = 44\%$

Q8. When SP is same: Net loss% = $R^2/100 = 625/100 = 6.25\%$

Q9. Gain% = $(\text{True} - \text{False})/\text{False} \times 100 = (1000 - 800)/800 \times 100 = 25\%$

Q10. Average of first n natural numbers = $(n+1)/2 = 51/2 = 25.5$

Q11. Correct avg = $72 + (55 - 85)/30 = 72 - 1 = 71$

Q12. Weighted avg = $(16 \times 50.25 + 8 \times 45.15)/24 = (804 + 361.2)/24 = 48.55 \text{ kg}$

Q13. Circular permutation = $(n-1)! = 5! = 120$

Q14. LEADER: 6 letters, E repeated twice. Arrangements = $6!/2! = 360$

Q15. Total - No woman = $11C5 - 7C5 = 462 - 21 = 441$

Q16. $11!/(4! \times 4! \times 2!) = 34650$

Q17. Favorable outcomes = 6 [(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)]. $P = 6/36 = 1/6$

Q18. Favorable = $4+4 = 8$. $P = 8/52 = 2/13$

Q19. $P(\text{solved}) = 1 - P(\text{none}) = 1 - (1/2)(2/3)(3/4) = 1 - 1/4 = 3/4$

Q20. $366 = 52 \text{ weeks} + 2 \text{ extra days}$. Favorable = 2 cases out of 7. $P = 2/7$

Q21. Time = $ab/(a+b) = 150/25 = 6 \text{ days}$

Q22. $M1 \times D1 = M2 \times D2 \Rightarrow 12 \times 20 = 15 \times D2 \Rightarrow D2 = 16 \text{ days}$

Q23. Differences: 4, 6, 8, 10 (AP, diff=2). Next diff = 12. Answer = $30+12 = 42$

Q24. Gaps: +2, +3, +4, +5, +6. $P(16)+6 = V(22)$

Q25. Each letter is shifted +1: C+1=D, O+1=P, M+1=N, etc. So P+1=Q, R+1=S, I+1=J, N+1=O, T+1=U, E+1=F, R+1=S $\Rightarrow$ QJSJOUFS